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Research paper

Analysis of aggregated load consumption forecasting in short, medium and long term horizons using Dynamic Mode Decomposition

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ABSTRACT

In response to the growing need for precise energy demand forecasting in distribution grids, this study examines the effectiveness of the Dynamic Mode Decomposition (DMD) algorithm for short, medium, and long-term demand forecasting in distribution grids, leveraging diverse time series energy consumption data. The DMD algorithm, a data-driven approach renowned for its capacity to discern frequencies and trends, will be compared against established forecasting techniques, including XGBoosting Regressor, Long Short-Term Memory, and PROPHET. The research is conducted through a comprehensive case study comprising three distinct phases: study on pattern capturing of DMD, comparison in Long term horizon with all models studied and robustness study of DMD. It recalls upon real aggregated hourly energy consumption data sourced from a Spanish distribution grid. The practical applications and benefits of DMD in time series forecasting are examined carefully in order to show its comparative advantages over alternative forecasting models. To ensure a rigorous assessment of the findings, the analysis will employ four key statistical metrics: Root Mean Square Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). The results obtained with the case studies showed a 75% reduction on computational waste compared to other models and the results are closer to classical models, obtaining an improvement in the best case study of 44% comparing RMSE with Prophet. The insights gained can help future applications and research in the field.

1. Introduction

In recent years, the scientific and technical community has seen a significant increase in the techniques and accuracy of the forecasting of variables in the energy field through algorithms and models (Dylewsky et al., 2020; Kaggle, 2024). Many studies have utilized the introduction of Artificial Intelligence (AI), Machine learning (ML) and other models to analyze future requirements. However, one algorithm has attracted the interest of many researchers in the recent years because of its ability to capture patterns and complex trends using the Singular Value Decomposition (SVD). This algorithm is known as Dynamic Mode Decomposition (DMD) (Anon, 2024; Chen, 2021; Kaggle, 2024).

In the energy consumption context, load consumption and renewable energies are of paramount importance in the modern world. Aggregated electrical load consumption represents a significant part of energy demand and its accurate forecasting and management are crucial for grid stability and efficient resource allocation (Phuangpornpitak and Prommee, 2016; Desai et al., 2021). The use of ML based algorithms is also being applied for renewable generation forecasting, for instance photovoltaic and wind generation (Guo et al., 2020; Jung et al., 2020; Liew et al., 2022).

Load consumption forecasting provides clear benefits over various time horizons. In the short term, spanning hours to days, it assists with optimizing resource allocation and grid stability, thus ensuring a reliable and efficient energy supply (Institute of Electrical and Electronics Engineers, 2012). In the medium-term, covering weeks to months, it supports capacity planning and infrastructure management, thereby enabling cost-effective decisions and robust supply chain operations (Polytechnic Institute of Setúbal et al., 2020). However, load consumption forecasting is also key in the long-term horizon, spanning into years or even decades in the forthcoming times. In this regard, it serves as a crucial element in determining the appropriate grid reinforcements and derived investments, directing the shift towards sustainable and eco-friendly energy origins, and providing a durable, environmentally conscientious, and reliable energy system for forthcoming generations (Habbak et al., 2023; Saldaña-González et al., 2024; Hernandez-Matheus et al., 2024).

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For recent short-term forecast studies, the algorithm known as DMD has proven to be a powerful tool for obtaining fast and precise predictions compared to other ML and DL models (Tirunagari et al., 2017). This model offers numerous advantages when considered in contrast with other regression models in a short term horizon. It utilizes the decomposition algorithm, which significantly assists in reducing the matrix dimension of the utilized data by using only the representative eigenvalues of the captured patterns. Due to the advantages in a short term horizon forecasting, this algorithm has been lately widely used for the data scientists. In the studies (Mohan et al., 2018; Krake et al., 2021; Namas et al., 2021) the authors explain the incorporation of DMD for a short term horizon instead of other models using different improvements and modifications on the DMD algorithm is shown, and proved that works and fits tightly. Other studies, such as Dylewsky et al. (2020) and Revati et al. (2021) use Neural Networks and ML models to analyze short-term load consumption forecasting. In these studies the authors analyzed the performance of models such as Long Short-Term Memory (LSTM), Auto-Regressive Integrated Moving Average (ARIMA), or different variances of LSTM and Recurrent Neural Networks (RNN) in a short term horizon and compare with the test set to see how the improved models can perform the predictions. Eventually, in Gugaliya et al. (2022), the authors analyze wind and solar generation data at 5-min intervals to conduct a long-term study. The authors of the mentioned article use the term long-term to refer to a larger number of data but a shorter time span.

Other decomposition models within the same family, such as PCA or ICA, have demonstrated highly precise results for clustering methods. Similar to DMD, they capture temporal patterns and trends to establish relationships and cluster variables. Even though these algorithms do not use the trends and patterns for forecasting, these intrinsic features demonstrate the exceptional capability of decomposition models as algorithms. The contributions using these decomposition models are in Mbuli et al. (2020). In many other articles, authors highlight the benefits of DL and ML models, as in the following articles (Son et al., 2020; Bentéjac and Martínez-Muñoz, 2020; Almazrouee et al., 2020). These studies are also interesting because by delving deeper into the functioning of these models, we can experiment and observe the advantages of this innovative algorithm. In contrast to Gugaliya et al. (2022), this article examines the medium term scope, bearing in mind that our data originates from various years in February, thus extending our actual scope to 5 years hence. This nuanced interpretation of long-term strengthens the concept beyond what was previously considered.

The aim of this paper is to evaluate the performance of DMD under different time horizons, but unlike the articles mentioned above, focusing on mid and long term horizons. The authors will analyze under which circumstances this data-driven model can be used in a more precise manner compared to well-known forecasting techniques. For that purpose, we divided this study into two different analysis. The first proposed approach involves using DMD algorithm to forecast aggregated load consumption from substations datasets. The objective is to evaluate the performance of the algorithm under varying data conditions and identify which tendencies it captures most effectively. Subsequently, the authors will compare the DMD algorithm's accuracy against other models, in particular, XGBoosting Regressor (XGBoost), LSTM and PROPHET, based on error metrics: Root mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) to elucidate any potential advantages or differences in the models' characteristics. Finally, we will discuss scenarios where DMD is preferable due to the characteristics of the input data and in which horizons performs better.

The main contributions of this paper are (i) identification of the potential of applying DMD algorithm for aggregated demand forecasting in the long-term horizon and development of a DMD based demand forecasting model; (ii) comprehensive analysis of the performance of DMD for aggregated demand forecasting in short, medium and long-term horizons, compared to other approaches conducting an in-depth analysis in different scenarios to evaluate its versatility and advanced pattern recognition.

2. Dynamic Mode Decomposition - DMD

DMD is a data-driven algorithm used for analyzing and forecasting the behavior of complex dynamical systems (Revati et al., 2021). It extracts coherent spatio-temporal patterns, or modes, from high-dimensional data (Tirunagari et al., 2017). These modes capture the dominant underlying dynamics of the system, allowing us to understand its behavior and is capable to predict future patterns (Prasadan and Nadakuditi, 2019).

The governing equation for a dynamical system can be represented as (Prasadan and Nadakuditi, 2019):

$$\frac{dz}{dt} = f(z,t) \tag{1}$$

where $z \in \mathbb{R}^n$ is the vector comprising the system states. The data is further arranged into two matrices:

$$Z1 = \begin{bmatrix} | & | & & | \\ z_1 & z_2 & \cdots & z_{m-1} \\ | & | & & | \end{bmatrix} \tag{2}$$

$$Z2 = \begin{bmatrix} | & | & & | \\ z_2 & z_3 & \cdots & z_m \\ | & | & & | \end{bmatrix} \tag{3}$$

The best-fit matrix is computed as:

$$P = Z_2 Z_1^{\dagger} \tag{4}$$

With this matrix decomposition, we want to find a linear approximation of the non-linear dynamical system using:

$$Z_2 \approx P Z_1 \tag{5}$$

The DMD algorithm starts by capturing system state variable snapshots at varying time points. These snapshots are then utilized to create a matrix of stacked state vectors (X). The DMD algorithm decomposes this matrix into its Singular Value Decomposition (SVD), revealing the system's modes (Namas et al., 2021). These modes correspond to eigenvectors of the matrix, representing spatial patterns, and their corresponding eigenvalues indicate temporal dynamics (Baddoo et al., 2023).

Let X be the matrix of state vectors obtained from the system snapshots

$$X = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{bmatrix} \tag{6}$$

The SVD decomposition of this matrix is:

$$X = U \Sigma V^* \tag{7}$$

where: U is the left singular vector matrix, Σ is the diagonal matrix of singular values, V is the right singular vector matrix, and V^* denotes the conjugate transpose of V.

$$Z_1^{\dagger} = V \Sigma^{-1} U^* \tag{8}$$

These equations describe the SVD-based approach in the context of DMD, facilitating the extraction of meaningful modes from high-dimensional data (Giannakis, 2019). Fig. 1 shows DMD's construction and methodology of forecasting.

2.1. Advantages in forecasting

DMD presents several key advantages in the field of forecasting (Filho and Santos, 2019):

- Applicability to Diverse Systems: DMD can be applied to a wide range of systems without the need for prior knowledge of related equations.

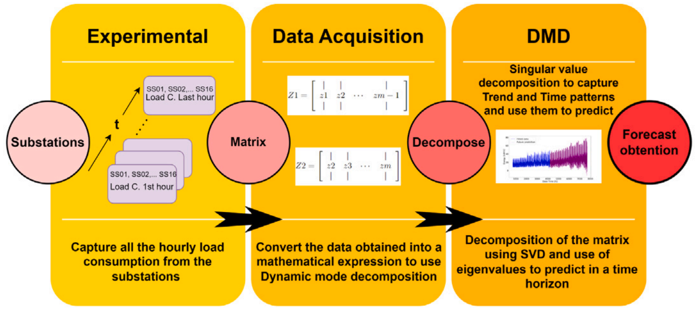

Fig. 1. DMD algorithm explanation of the process.

- Handling High-Dimensional and Nonlinear Data: It is capable of handling high-dimensional and nonlinear data, making it valuable for complex systems in areas such as fluid dynamics, neuroscience, and finance.
- Dimensionality Reduction: DMD delivers a reduced, low-dimensional approximation of the system, leading to a reduction in computational complexity and a more efficient analysis.

However, some previous works on DMD forecasting have stated limitations, for example in Liew et al. (2022), the authors discuss the main limitations of the model in this particular forecasting scenario. It presupposes linear system dynamics, which may not always be applicable in highly nonlinear systems. Moreover, uniform temporal sampling of data snapshots is assumed, although deviations or irregularities in the sampling rate may affect the precision of results. Additionally, DMD requires the data snapshots to cover the entire range of dynamics, and any limitations or biases in the data can impact the quality of the decomposition.

2.2. Hyperparameter tuning in DMD

In this DMD algorithm implementation, hyperparameters are tuned by adjusting both the threshold (thresh) parameter and the time series length (ts_length) (Krake et al., 2021).

The thresh parameter is employed in the singular value decomposition (svd_x1 method) to establish the rank of decomposed matrices, which effectively controls the truncation of singular values. Altering this threshold variable enables one to regulate the number of preserved modes in the DMD analysis. To determine the optimal value for this parameter, a grid search was conducted using multiple values, resulting in the identification of the best option among various choices.

The hyperparameter ts_length is set to 80% of the total length of the training set. This allows the algorithm to predict future values over a longer period of time. This parameter is critical to the algorithm as it determines the number of data points in the training set used for vector decomposition. It is not possible to use 100% of the training values due to potential dimensional errors, so a lower percentage is needed to fit the train set correctly (Kong et al., 2020). This hyperparameter depends on the data distribution and its tuning can be done through methods like grid search or Bayesian search.

3. Methodology of the construction of the models

For the creation of all the models used in this article, the same procedure has been realized for obtaining the best results possible. Hyper-parameter tuning is a crucial step in optimizing machine learning models like DMD, Prophet, LSTM, and XGBoost. It is essential to enhance model accuracy and ensure it performs well on new data.

Here's how it works:

(1) **Select Parameters**: Identify the critical settings (hyperparameters) for your model.
(2) **Create a Grid of Parameters**: Define a range of values to test for each hyperparameter.
(3) **Train & Evaluate**: The model is trained with various combinations of these values and evaluated on its performance.
(4) **Best Settings**: Choose the combination of values that results in the best performance.

This process prevents under-fitting and overfitting, making the model more reliable, efficient, and adaptable. Fig. 2 shows a flowchart with the methodology of the construction for each model, also shows the iterative process over the hyperparameter tuning and constant evaluation to obtain optimal models.

The hyperparameter tuning through grid search (or similar methods) is an essential step in parameter optimization for machine learning forecasting models. It ensures that the model is fine-tuned to deliver the best performance of the model. This process is an important component of the broader flowchart that guides data analysis and prediction tasks. The statistical metrics used for the numerical analysis of the performance of the models are the next ones, specifying that, n states for every time step, y the actual value of the test set for the n time step and $\hat{y}_i$ for the predicted value of that specific time step:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{9}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{10}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{11}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{12}$$

4. Case study

4.1. Data description

This study examines the load consumption patterns of various transformers within a distribution grid located in Granollers, Catalonia, Spain. The dataset includes hourly load consumption data from 16 substations in the distribution grid. The dataset focuses on Februaries, the coldest month of the year, known for its peak residential load consumption. It spans from February 2012 to February 2019 and includes 5377 cleaned and validated hourly records from 16 substations. This comprehensive dataset offers valuable insights into load consumption trends over a seven-year period, allowing for a detailed analysis of consumption patterns during the peak demand month.

Fig. 3 shows the data available for a specific substation in the distribution network, along with a regression line indicating the increasing trend in load consumption over time. A zoomed-in view of a week in February is provided to better illustrate daily consumption patterns. Identifying trends in electrical demand is essential for effective long-term forecasts.

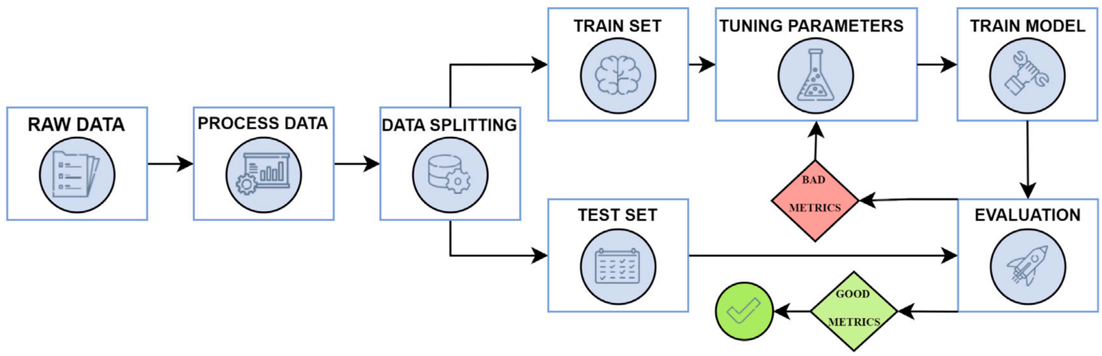

Fig. 2. Flowchart of the creation of the models.

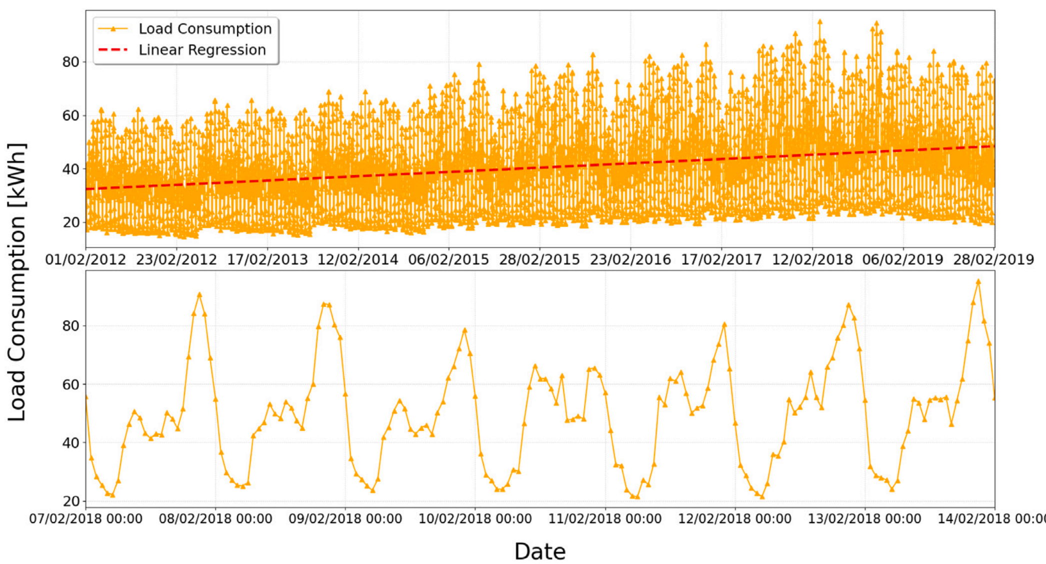

Fig. 3. Top figure: Data available of the SS06; Inferior figure: Zoomed area of some day's load consumption.

In conventional forecasting ML and DL models, data is typically split into an 80% train set and a 20% test set for evaluation. However, the DMD algorithm utilizes a different approach. As discussed in Section 1, this study evaluates predictions across various horizons. Specifically, the train set spans approximately 7 years of February data, predicting the next 50 days, or around 2 years ahead. Referring to Phuangpornpitak and Prommee (2016), short-term spans 168 h (7 days), medium-term spans from 158 h to 672 h (1 month), and long-term spans from 672 h to 3500 h (roughly 5 months, equivalent to 5 Februaries). In summary, the model utilizes the last 3500 values for prediction, incorporating recent and relevant data points. Given the extensive dataset used to fit the DMD model, it can forecast the next 145 days (approximately 5 months). This approach facilitates identifying the best fits of the algorithm across different substations.

After thorough examination, no anomalies were found regarding missing or null values. Consequently, the preprocessing of the data demonstrates that the load consumption dataset is highly accurate. Also, for lag features consideration on long-term DMD forecasting, despite data stationarity, it is advisable to avoid lag features due to complexity concerns, diminishing returns and speed correlation decay. Lag features increase model complexity especially for understanding dominant patterns (Yang et al., 2022). Finally, with 80% of the training data as fitted data, the inclusion of lag features becomes unnecessary.

In this paper, we analyze and compare three substations from our dataset — SS02, SS06, and SS12 — using classical models across different case studies. For a deeper understanding of the data, Fig. 4 presents box-plots illustrating the distribution of the data we will use. We acknowledge the presence of outliers in SS06 and SS12, which are utilized in Section 4.3. These outliers originate from the original data and may introduce some dispersion when evaluating the metrics. However, they do not significantly affect the accuracy of the models.

Exogenous variables that show high correlation with the target variable could be used to improve the accuracy of the forecast. For instance, weather variables could be considered in the model for its potential effect on the energy demand. To evaluate this, Fig. 5 shows a correlation heatmap that displays the existing correlation between temperature (°C), pressure (Pa), UV index, precipitation (ml/m^2) and energy consumption at the grid operated by the DSO on the exact day-time and GIS location. The weather conditions were extracted using the NASA Power Interface (NASA, 2024).

Fig. 5 shows that the surface temperature has the highest correlation index of 0.4043 with the load consumption pattern. It is important to note that correlation indexes range from 1 to −1, with 1 being the most correlated 0 the least correlated and −1 being the most inverse correlated. As temperature forecasts typically exist for 1 or 2 weeks ahead, it would not be realistic to include this feature in the model for long-term forecasting because it may increase the uncertainties and errors of the predictions (Lusis et al., 2017). Also, exogenous variables were excluded from the following case studies because the aim of this work is to evaluate the performance of DMD over different time horizons, rather than comparing different DMD variants with exogenous variables, such as DMD with Control described in Habbak et al. (2023). Finally, authors in Qiu et al. (2021) and Fortes et al. (2015) suggest to consider for medium and long-term forecasting, socio-economic factors such as population growth, economic growth, etc.

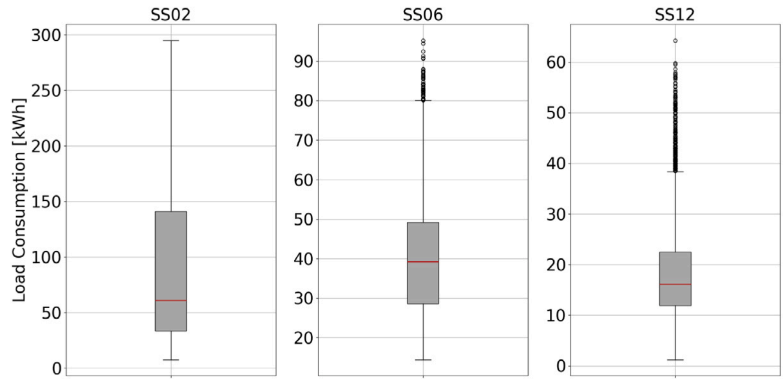

Fig. 4. Box-plot of the original data from SS02, SS06 and SS12.

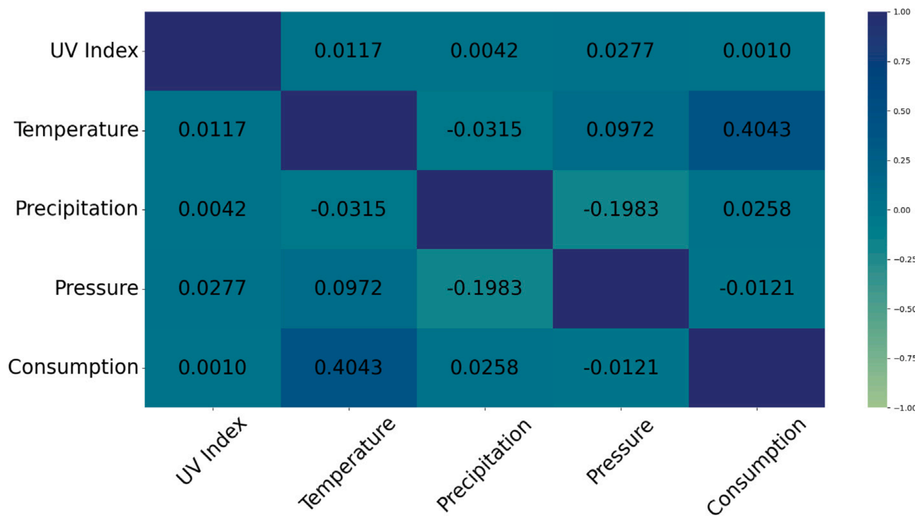

Fig. 5. Heatmap correlation plot.

4.2. DMD forecasting

With the use of the DMD algorithm explained in Section 2 in this case study we will provide the forecast of all the substations in the network in order to visualize the performance of the model for different tendencies or data structures. For this particular case study and through trial and error with test validation, it is been found that using 80% of the training data consistently produces reliable and accurate results. The construction of this particular model contemplates the ***Central Limit Theorem***(CLT), where each regressor should have at least 30 observations (Brownlee, 2020).

In addition, we evaluate the capacity to discern temporal patterns and observe the trend of the data, so as to establish and validate any potential advantages over other models. We will then forecast roughly 3500 values, which are evidently beyond the range of the test set incorporating 1176 values.

Table 1 shows the error metrics used to evaluate the performance of the DMD algorithm into different sets of data for the test length, set as 1176 h ahead which corresponds to approximately 50 days ahead in the future, a medium-term study according to the length of the prediction.

Using the *"Composite Score Method"*, which using all the information for metrics makes a single score of the performance to see which are the most accurate ones, the four best ones are in descending order: **SS06**, **SS13**, **SS09** and **SS12**. Fig. 6 show the predictions made with DMD algorithm compared to the test set and a zoomed area of the last 10 days potentially useful to observe the great adjustment of the algorithm for the **SS06** (the best according to the composite score).

Table 1
Metrics using DMD for the longest horizon of 1176 h all the substations.

Substation	MAE (kWh)	MAPE (%)	RMSE (kWh)	MSE (kWh^2)
SS01	8.66	20.52	10.44	109.08
SS02	21.09	27.56	30.31	918.7
SS03	10.32	14.97	13.73	188.70
SS04	10.39	17.54	14.33	205.35
SS05	11.62	22.95	15.81	250.13
SS06	5.70	13.04	7.36	54.29
SS07	12.43	25.98	16.35	267.53
SS08	9.87	25.10	11.91	142.05
SS09	4.01	35.84	4.86	23.65
SS10	7.68	46.78	10.35	107.29
SS11	10.15	22.86	12.85	165.14
SS12	4.61	27.68	6.19	38.39
SS13	2.07	37.15	2.68	7.19
SS14	10.84	9.74	13.64	186.11
SS15	17.55	37.03	22.34	499.25
SS16	9.05	51.49	11.91	141.93

As evident from Fig. 6, this model effectively conforms to the time-based patterns recurrent in the dataset to accommodate diverse consumption peaks. As far as the second objective of the study is concerned, in the next pictures, the predictions following the train data used can be seen for the best four models, and can be visible the capacity of this specific algorithm to capture the tendency patterns over time. The algorithm's ability to maintain consistent and linear predictions over a long time horizon is highlighted in Fig. 7 with

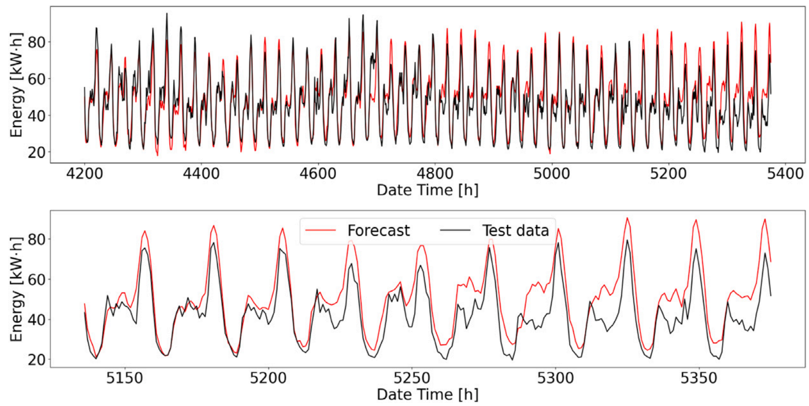

Fig. 6. Actual vs. Predicted energy consumption comparison - Zoom last 10 days.

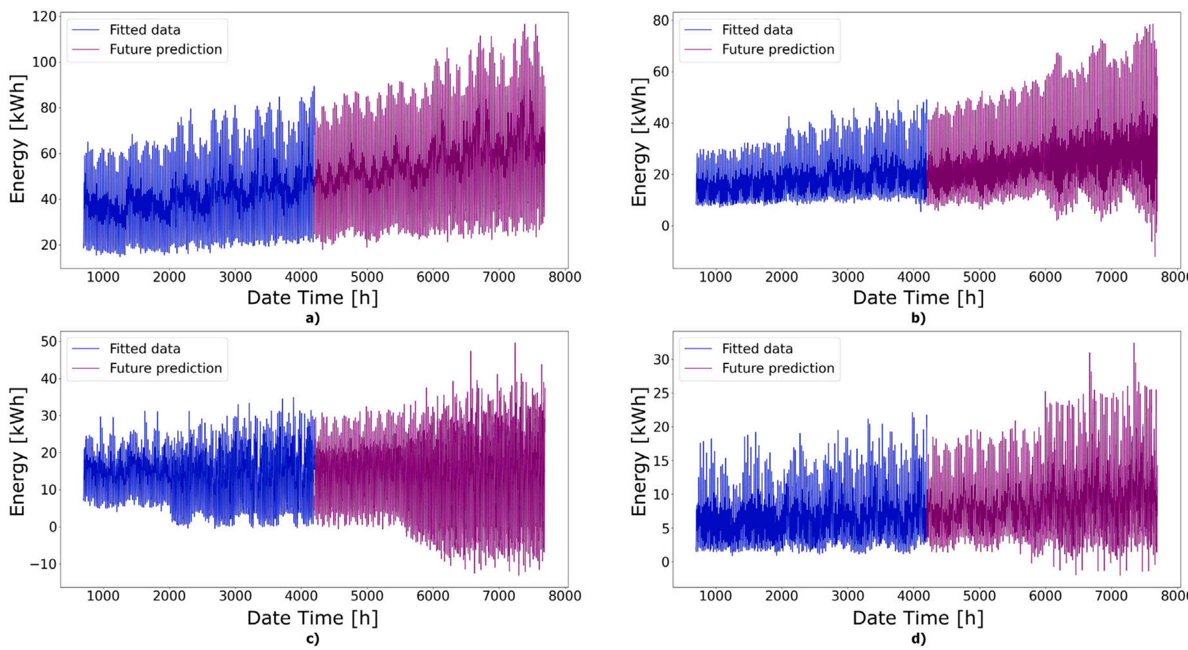

Fig. 7. Forecast out of range using DMD on (a) SS06, (b) SS09, (c) SS12 and (d) SS13.

four substation's performance. This feature is one of the model's clear advantages.

The prediction results from the DMD model, visible in Fig. 7, are able to capture the trend in short and medium term horizon (several months), combining these simplest patterns with a tendency pattern. This can be very useful in order to observe a possible tendency in the future to adapt the network in a planning study. By analyzing this load consumption data, we aim to gain a deeper understanding of each substation performance in the medium-term. This knowledge will aid in optimizing power distribution and enhancing the overall efficiency and reliability of the electrical network in Granollers, ultimately contributing to more sustainable and resilient energy management. It is important to remark that some actions of post-processing can be needed for this kind of studies. For example in Fig. 7(c) there are negative values that are nonsense knowing that we are dealing with load consumption values, so after the predictions are made we should set a minimum value as 0 kW h.

4.3. DMD forecasting compared with XGBoost, LSTM and Prophet forecasting

Once the DMD algorithm's predictions have been obtained, the second step will involve contrasting them with the forecasts of other models in two of the substations. The fairness of the comparison is ensured by the fact that all the models tested are equally trained in the same circumstances and only predict in the test set of the specific substation "**SS06**". So the horizons analyzed and the input data is the same for all of them.

In order to make a deeper study, a comparison is made between the forecast of a Machine Learning model **XGBoosting Regressor**, a Deep Learning model, **LSTM** model and a Statistical Model, **Prophet**. With these algorithms a more exhaustive comparative can be achieved (Cordeiro-Costas et al., 2023). The construction of these algorithms also contemplates the CLT explained in Section 4.2.

The substations studied are **SS06** and **SS12**. The purpose of selecting these substations is to enable a comprehensive analysis by contrasting the outcomes of the best-performing substation with that of the fourth-best one. The substations examined in the second case study are the top-performing station and the fourth-best one according to the Composite Score Test mentioned in Section 4.2.

4.3.1. XGBoost regressor

XGBoost (Extreme Gradient Boosting) is a gradient boosting algorithm that has gained popularity due to its performance in regression and classification tasks. It works by iteratively adding weak learners, typically decision trees, to form a strong predictive model. Each new tree is built to correct the errors made by the previous trees, gradually improving the model's predictive capability (Bentéjac and Martínez-Muñoz, 2020; Guo et al., 2020).

In this study the XGBoost model is applied to a regression problem. The data is divided into train and test sets, and a scaler is used to normalize the data. The XGBoost model is initialized, and a parameter grid is defined for grid search. Grid search is performed using the GridSearchCV function, which exhaustively searches through the parameter grid to find the combination of hyperparameters that minimizes the mean squared error (MSE) on the test set.

The best hyperparameters obtained from the grid search are selected and used to make predictions on the test set. The predictions are then inverse transformed to the original scale using the scaler. The performance of the model on the test set is evaluated using various metrics such as mean squared error (MSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and root mean squared error (RMSE). Results from the best model configuration are evaluated based on the error metric scores.

4.3.2. LSTM model

The LSTM model is a type of recurrent neural network (RNN) that is effective for sequence data analysis and prediction. It consists of multiple LSTM layers followed by dropout layers for regularization and a final dense layer for output (Goodfellow et al., 2016). The LSTM architecture includes cell states and various gates (input, forget, and output gates) that help in regulating the flow of information, thus allowing the model to maintain long-term dependencies and mitigate the vanishing gradient problem typical in traditional RNNs.

The activation function used in the LSTM layers is typically the hyperbolic tangent (tanh) for the cell state and sigmoid for the gates. However, ReLU (Rectified Linear Unit) can also be used in certain configurations. The optimizer used for compiling the model is Adam, which combines the advantages of two other extensions of stochastic gradient descent, namely AdaGrad and RMSProp, providing efficient and reliable training (Revati et al., 2021). The model is trained using the Mean Squared Error (MSE) loss function, which is standard for regression tasks (Son et al., 2020).

The LSTM model is particularly powerful for time series forecasting tasks due to its ability to learn and leverage temporal patterns in the data (Lotfipoor et al., 2024). It can capture seasonality, trends, and complex dependencies, making it valuable for forecasting tasks. The model's architecture allows it to remember important information over long sequences, which is crucial for accurate time series predictions. Moreover, LSTM networks can be stacked, allowing for deeper models that can learn more abstract representations of the input data, thus enhancing their predictive performance.

4.3.3. Prophet

Prophet is a robust time series forecasting method developed by Facebook's Core Data Science team. It is designed for forecasting with daily observations and is particularly well-suited for business or economic time series data. This method excels at handling real-world time series data and has several advantages (Almazrouee et al., 2020).

At its core, the Prophet model comprises several key components (Desai et al., 2021). It excels in capturing the underlying data trends, recognizing recurring seasonal patterns, and accommodating the influence of holidays.

Prophet decomposes a time series into three main components: **Trend**, **Seasonality**, and **Holidays**.

- The **Trend** is represented as a piece-wise linear growth curve, providing adaptability in modeling data trends. It considers factors like growth rate, adjustment rate, offset, and trend changepoints. Analysts can manually adjust these changepoints to capture significant events in the time series.
- **Seasonality** captures recurring patterns that may occur daily, weekly, or annually using the Fourier series, offering a nuanced understanding of load consumption variations over time.
- **Holidays** introduce unique variations in load consumption. Users can specify a list of holiday dates, and the model explicitly accounts for their effects. A smoothing parameter (κ) allows for the customization of holiday impact.

Table 2
Optimal hyperparameters after the grid search for the data used for the different algorithms.

Algorithm	Hyperparameter	Value
XGBoosting	Number of estimators	150
	Learning rate	0.01
	Eta	0.3
LSTM	Number of LSTM layers	2
	N° of neurons in LSTM layers	64, 64
	N° of neurons in dense layer	128
	Dropout rate	0.2
	Optimizer	Adam
	Batch size	32
	Number of epochs	50
PROPHET	Seasonality mode	Multiplicative
	Changepoint prior scale	0.1
	Daily seasonality	True
	Yearly seasonality	False
	Interval width	0.95
	Custom monthly seasonality	Period: 672, Fourier Order: 3
	Holidays	False

One of Prophet's strengths is its ability to handle missing data and outliers without extensive preprocessing. It can impute missing data in the observed time series. Prophet provides a measure of uncertainty for each forecast point, quantifying prediction intervals for informed decisions. It also accommodates additive and multiplicative seasonality and allows users to specify the forecast horizon. The method can automatically detect change points, representing abrupt shifts in the trend. It employs a Bayesian framework for robust component fitting and uncertainty estimation.

In summary, Prophet is a user-friendly and robust method for time series forecasting, particularly valuable for real-world data with strong seasonal and holiday effects; also, its ability to handle missing data, provide uncertainty estimates. However, the choice of forecasting method should depend on specific data characteristics and analysis objectives.

4.3.4. Optimal hyperparameters for the models

Once the grid search procedure defined in Section 3 is done for the models studied, the optimal parameters can be seen in Table 2 for the different models and for this particular data. It is important to remark that this procedure, depending on the grid created can be computationally expensive if the number of combinations is huge.

Once we have the models constructed, we can now evaluate the performance of every one of them and compare them with DMD. Section 4.3.5 show the comparison in a long term horizon for two different substations.

4.3.5. Models comparison

- ***Substation 06***

Once the predictions are obtained, the statistical metrics used are show in Table 3, just for the largest time horizon of prediction, in order to observe the performance in overall the models used. As can be seen, the behavior between models is similar, observing that DMD has lower RMSE for example than PROPHET.

Fig. 8 displays a comparison of all metrics among the models, demonstrating their evolution over time. The results reveal that all the

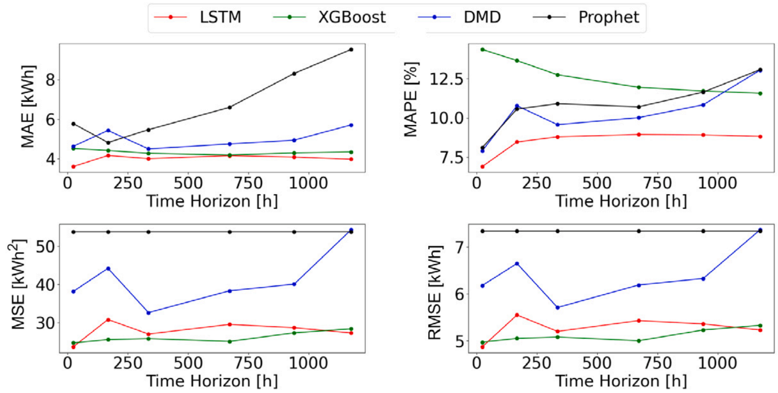

Fig. 8. Metrics evolution in short, medium and long-term in SS06.

Table 3
Metrics for the longest horizon of 1176 h in SS06.

Algorithm	MAE (kWh)	MAPE (%)	RMSE (kWh)	MSE (kWh^2)
XGBoosting	4.35	11.56	5.33	28.41
DMD	5.70	13.04	7.36	54.29
LSTM	3.98	8.80	5.22	27.34
Prophet	8.59	13.27	7.61	55.4

Table 4
Metrics for the longest horizon of 1176 h on SS12.

Algorithm	MAE (kWh)	MAPE (%)	RMSE (kWh)	MSE (kWh^2)
XGBoosting	3.71	24.08	4.91	24.11
DMD	4.61	27.68	6.19	38.39
LSTM	4.09	26.57	5.31	28.25
Prophet	9.62	34.37	11.22	129.39

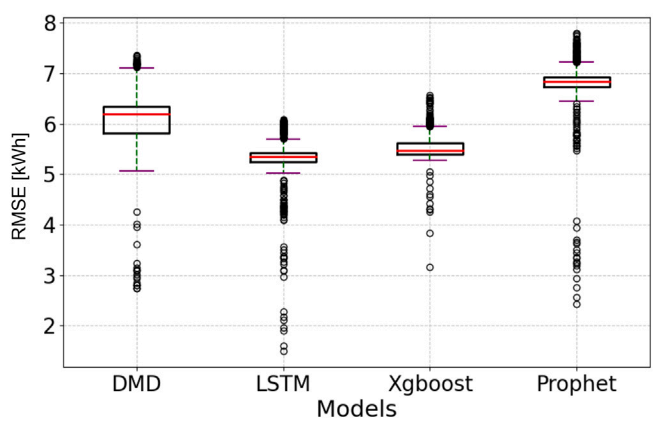

Fig. 9. Box-plot of the RMSE statistical distribution in SS06.

models performed exceptionally closely ranked. Using these different time horizons, we can analyze the evolution of the metrics on short, medium and long-term horizons, acknowledging the aim of this paper. As this figure shows, DMD shows accurate metrics when compared to other models in all the horizons, and even though metrics are not as good as LSTM or XGBoost in long term, the evolution is not far.

This performance is visible in the box-plot of the RMSE and its distribution across step time intervals as shown in Fig. 9. Using the RMSE distribution can be seen the similarities on good performance of the models, also observing the better performance of the DMD algorithm than Prophet. As shown in the box plot below, the distribution of the RMSE (Root Mean Square Error) metric reveals the presence of outliers. As explained in Section 4.1, these outliers can come from the original data distribution and can have a significant impact on the RMSE value over time because the RMSE is sensitive to large deviations; it squares these errors, amplifying the impact of the outliers. Despite the presence of these outliers, the final RMSE value indicates that the performance of the model is generally good, suggesting that it fits the data well. Although the outliers contribute to the RMSE, the overall accuracy of the model remains high, demonstrating that it can make reliable predictions for the majority of data points.

For further analysis and better visual comparison of the model's performance Fig. 10 shows the last 5 days of the test set compared with all the predictions for XGBoosting Regressor, LSTM, DMD algorithm and Prophet. As can be seen, all the models are fitting correctly and closely to the actual values and DMD is predicting similar to the other models. This figure shows the capabilities of the models to adjust the peak and off-peak values, showing that DMD has more accuracy on the prediction of the both of them.

To a better explanation of the adjustment that these models are predicting, Fig. 11 shows the Scatter-plot representation comparing the predictions with the test set. As can be observed all of the models have an acceptable dispersion, and also assure there are not overfitting. Despite this fact, DMD shows a better dispersion to conclude the results are pretty accurate knowing that we are talking about a mid-term forecast. Fig. 11 also shows how XGBoost display tight clustering around the red line, indicating high accuracy with minimal deviation from outliers. LSTM and DMD has a moderate spread, demonstrating robustness but some sensitivity to higher values. Prophet shows a wider spread, indicating a significant impact from outliers and less accuracy. The dispersion observed can be attributed to the inherent anomalies present in the dataset.

- ***Substation 12***

To make a more exhaustive analysis of the complexity and performance of the DMD algorithm, the same study will be realized in another substation. In particular, chasing the ranking of DMD adjustment in Section 4.2, the fourth better performance of DMD is **SS12**. In this final study we want to show, that even though is not the best performance of DMD overall the sub-stations, the results are not so far to the other models, in some cases, even better. Table 4, compares the performance of the XGBoost, DMD, LSTM and Prophet models over 1176 h in SS12 using the MAE, MAPE, RMSE and MSE metrics. XGBoost shows superior accuracy with the lowest errors across all metrics, demonstrating its reliability and robustness in dealing with data variability. DMD, while competitive, shows higher error metrics, reflecting moderate sensitivity to outliers. LSTM performs better than DMD on MAE and RMSE, indicating effective handling of complex patterns, but is still affected by anomalies. Prophet has the highest errors, indicating significant sensitivity to outliers and lower accuracy.

In this substation, the non-linearity of the data affects directly onto the model's performance. However, between this model and alternative

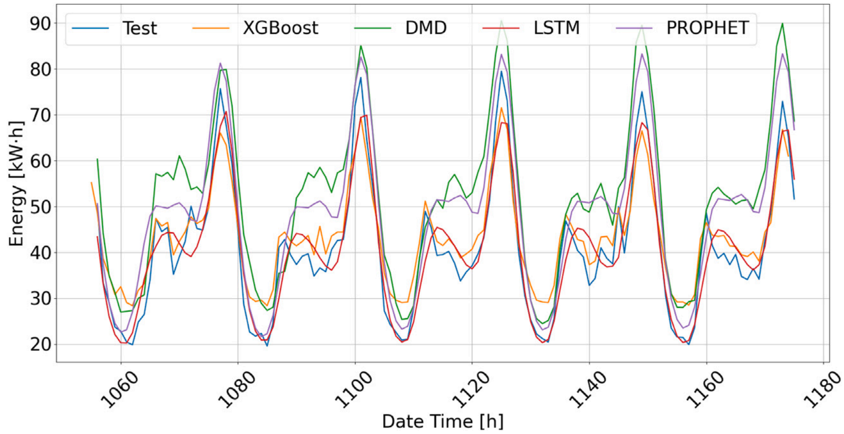

Fig. 10. Comparison of predicted value vs. real value (test) for different models in SS06.

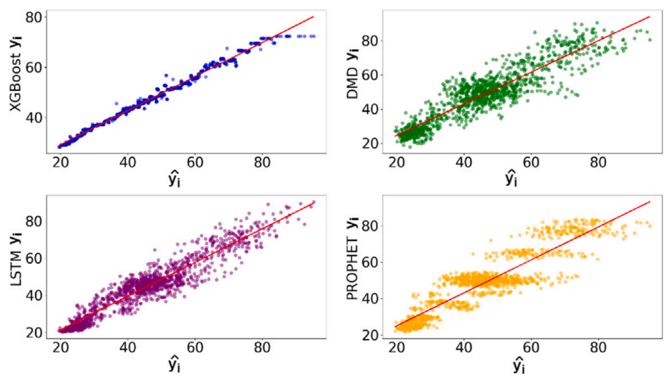

Fig. 11. Scatter-plots to see the adjustments of the predictions in SS06.

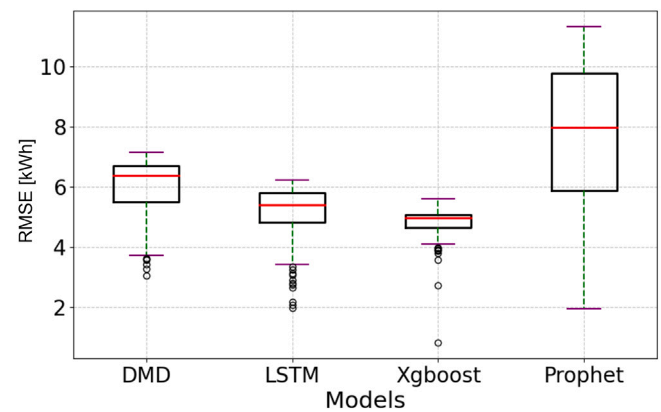

Fig. 13. Box-plot of the RMSE statistical distribution in SS12.

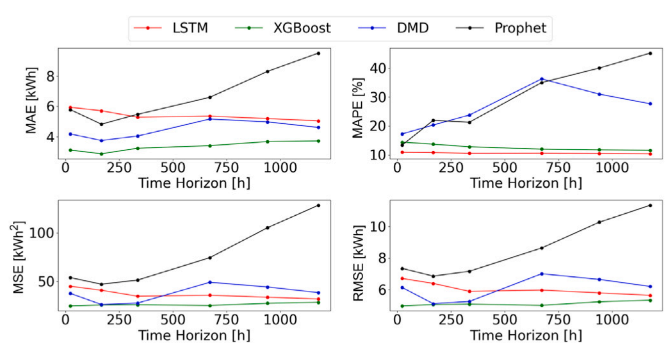

Fig. 12. Metrics evolution in short, medium and long-term in SS12.

ones, such as XGBoosting Regressor or LSTM, is not significant, in what error metrics represents. In contrast, PROPHET's performance is comparatively poor when dealing with such non-linear data. Fig. 12 shows the evolution of the metrics above time.

As observed in the previous substation analysis, Fig. 13 displays the Box-plot distribution of RMSE. Similar to the explanation provided earlier regarding the presence of outliers in this visualization, it is worth noting the discussion on data dispersion outlined in Section 4.1. Notably, in this particular substation, the presence of outliers in the original dataset was significant.

Fig. 14 shows DMD over-performing compared to XGBoosting and LSTM and Prophet presents the poorest performance. As can be observed for this data distribution in this particular substation, PROPHET has difficulties adapting the patterns while DMD, XGBoosting Regressor and LSTM show great adaptability and accuracy. DMD shows great capability of predicting near the test values even though the data is not as constant as in SS06.

As in the first study case, Fig. 15 shows the regression lines and the dispersion of the predictions for every model. The dispersion in this study es very alike to the one mentioned in Section 4.2. These scatter-plots assure that the models are not over-fitting. Also, they show that in this substation, PROPHET model shows a worse adjustment than the other models. Fig. 15 shows the predictions for SS12, revealing different levels of model accuracy and sensitivity to moderate outliers. XGBoost demonstrates the highest accuracy with tight clustering around the red line. DMD shows a broader spread, indicating moderate sensitivity to outliers, while LSTM exhibits a reasonable clustering but significant deviation at higher values. Prophet displays the widest spread, indicating substantial impact from outliers and the least accurate predictions.

To conclude, during all the study, the time of simulation also has been studied as a key advantage of DMD over the other models used. Its accuracy of pattern recognition in a computational speed highly superior than LSTM or Prophet.

The simulation time of the models analyzed in this paper were measured in order to observe the advantages in terms of computational efficiency of DMD. Table 5 shows the results in minutes for SS06 and SS12. This time is only for the prediction process after the hyperparameter tuning. Table 5 shows the computational advantages of DMD over LSTM, XGBoost and PROPHET, with an improvement in computational speed of 75% for LSTM and 47% for XGBoost for these cases.

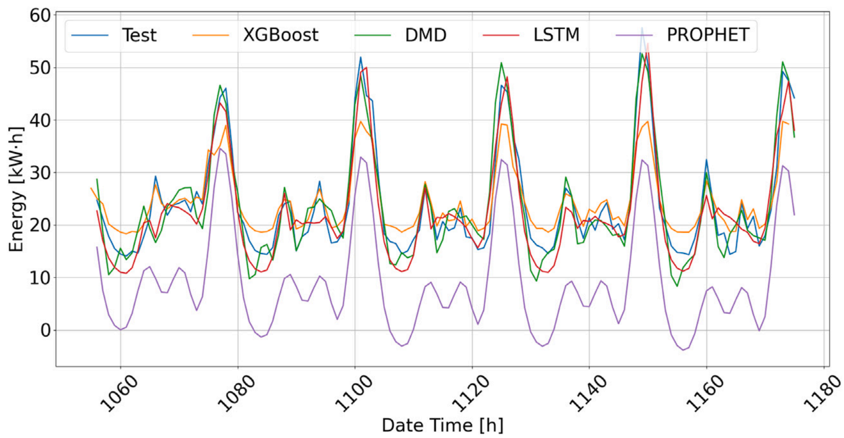

Fig. 14. Comparison of predicted value vs. real value (test) for different models in SS12.

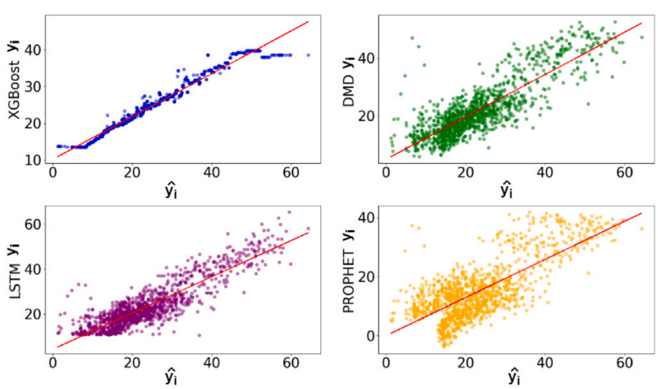

Fig. 15. Scatter-plots of the predictions in SS12.

Table 5
Time of simulation for all models tested in long term forecasting.

Algorithm	Time SS06 [min]	Time S12 [min]
XGBoosting	6.18	5.89
DMD	3.26	3.76
LSTM	13.11	15.8
Prophet	9.45	7.26

4.4. Testing DMD's robustness with weekend and week days splitting

As stated in Section 1, the aim of this paper is to show the advantages and complex pattern capturing of the DMD algorithm on large datasets. According to the previous case studies, the model shows outstanding results not only in terms of computational speed and complexity, but also in terms of pattern and trend detection with only two hyperparameters. The purpose of this section is to evaluate the impact of separating weekend and week days on the performance of the different forecasting models splitting weekend and weekday data to explore whether smaller data subsets, combined with merged predictions, yield improved results across all models under examination (Lusis et al., 2017).

The procedure of the study is similar as in Section 4.3 but in this case study, instead of using all the aggregated energy consumption's dataset the authors separated the data into two datasets, one dataset contains all the weekend days and another only with the week days. Using the same models as in Section 4.3 we trained the algorithms splitting the datasets into week and weekends separately in order to evaluate the performance combining the predictions over all the test set at once. This study addresses the significance of distinguishing between weekday and weekend patterns, potentially aiding models in capturing nuanced variations. However, despite we are aware of the DSO's location, uncertainty persists regarding the connection of each substation — whether for industrial or residential use — which may mitigate discrepancies between weekday and weekend patterns. Fig. 16 illustrates the patterns of a given week across three distinct substations.

For this study the substations studied will be SS06 in order to test the DMD model performance with this split approach, and finally SS02, which had the poorest performance over all the substations analyzed in Section 4.2. Using this methodology will lead us into the conclusion of robustness of DMD. Fig. 17 contains an outline of how the methodology explained above.

4.4.1. Substation 06

After splitting the data as mentioned above, the models perform worse due to the smaller amount of training data for some of the models. However, in DMD shows better results in capturing the time series growth trends. Once again, the superiority of DMD over other models is demonstrated by its ability to accurately capture temporal patterns with minimal training data, as seen in the case of weekends, without requiring any splits. At last, Fig. 18 shows one week compared to the predictions of all the models using this methodology. These figure shows the same week separating in colors weekdays and weekend days compared to the different models used.

4.4.2. Substation 02

In this case, the long-term DMD metrics obtained in Section 4.2 show that despite the more disparate data and complicated trends, the forecasting trend is very similar to the previous one. The models perform worse when the data is split due to the reduction of data to predict. However, DMD once again demonstrates that it can capture temporal patterns without requiring data with a lot of trend or repetitive patterns, nor does it need data separations. Also, Fig. 19 shows the comparison of all the models compared to the test data for one week as in the substation before. In this specific substation, the models accurately predict weekdays, with DMD showing superior performance in forecasting peak and off-peak hours. However, during weekends, the accuracy of peak hour predictions diminishes, primarily due to the disparate behavior observed between Saturday and Sunday. This discrepancy may be attributed to the substation's connections, potentially linking to residential areas or businesses that experience closures on Sundays. As depicted in Fig. 19, energy consumption notably decreases during this period.

Table 6 shows RMSE (kWh) values for four algorithms (XGBoosting, DMD, LSTM, Prophet) applied to two datasets (SS06, SS02) with data split into week and weekend days. In SS06, XGBoosting and LSTM perform best on the original data (5.33, 5.22 kWh) but worsen with splitting (6.63, 8.21 kWh). DMD shows moderate RMSE increases (7.36

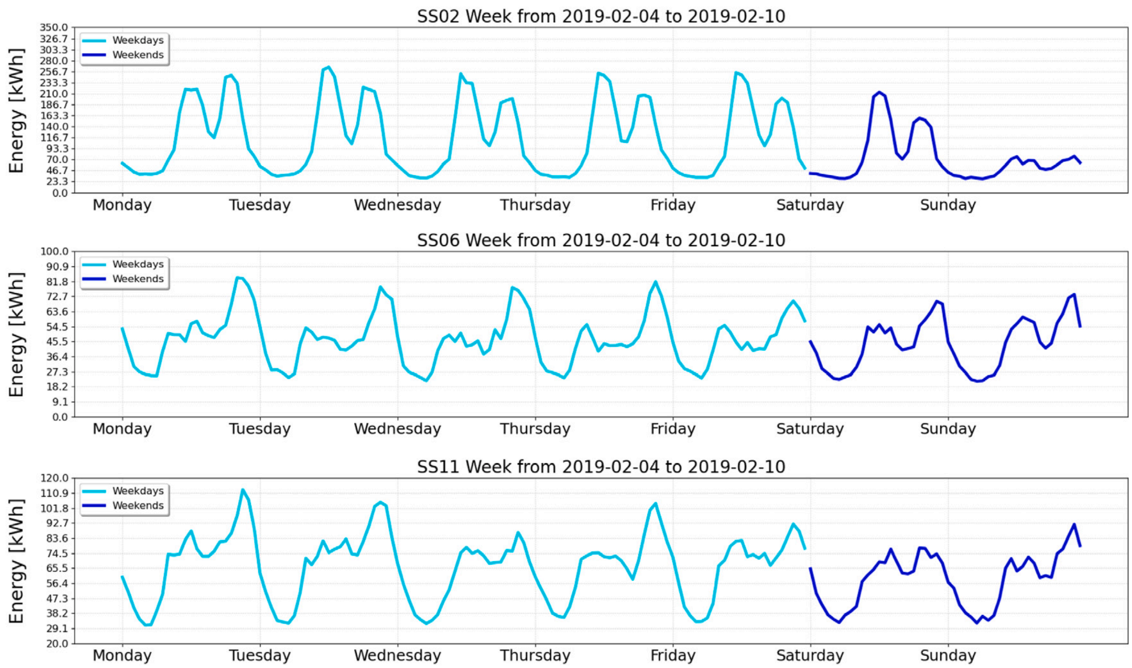

Fig. 16. Weekday and weekend difference in SS02, SS06 and SS11.

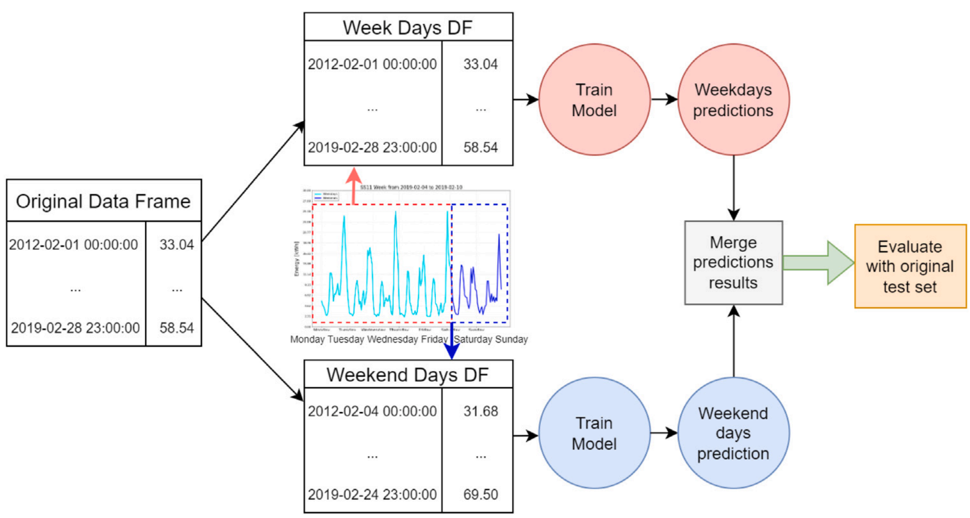

Fig. 17. Methodology splitting the original data set into week and weekend days.

Table 6
Metrics for the longest horizon of 1176 h in SS06 and SS02 splitting into week and weekend days.

	SS06			SS02		
	Algorithm	Original DF	Splitting	Algorithm	Original DF	Splitting
RMSE (kWh)	XGBoosting	5.33	6.63	XGBoosting	13.92	26.30
	DMD	7.36	8.39	DMD	30.38	32.73
	LSTM	5.22	8.21	LSTM	22.17	28.07
	Prophet	7.33	15.87	Prophet	30.23	37.98

to 8.39 kWh). Prophet is highly sensitive to splitting (7.33 to 15.87 kWh). In SS02, XGBoosting's RMSE significantly increases (13.92 to 26.30 kWh), while DMD shows stability (30.38 to 32.73 kWh). LSTM and Prophet both show considerable increases (22.17 to 28.07 kWh and 30.23 to 37.98 kWh, respectively). XGBoost shows the lowest RMSE for both the original data (13.92 kWh) and the split data (26.30 kWh), demonstrating its robustness and high accuracy in both scenarios. DMD has higher RMSE values (30.38 kWh for original data and 32.73 kWh for split data), suggesting that it is more sensitive to data splitting, leading to less accurate predictions.

LSTM performs better than DMD, but still shows a significant increase in RMSE when data is split (22.17 kWh for original data and 28.07 kWh for split data), indicating that while it handles complex patterns well, its accuracy decreases with data splitting. Prophet, with the highest RMSE values (30.23 kWh for original data and 37.98 kWh for split data), is the most affected by data splitting, reflecting its high sensitivity to changes in data structure and its lower overall accuracy.

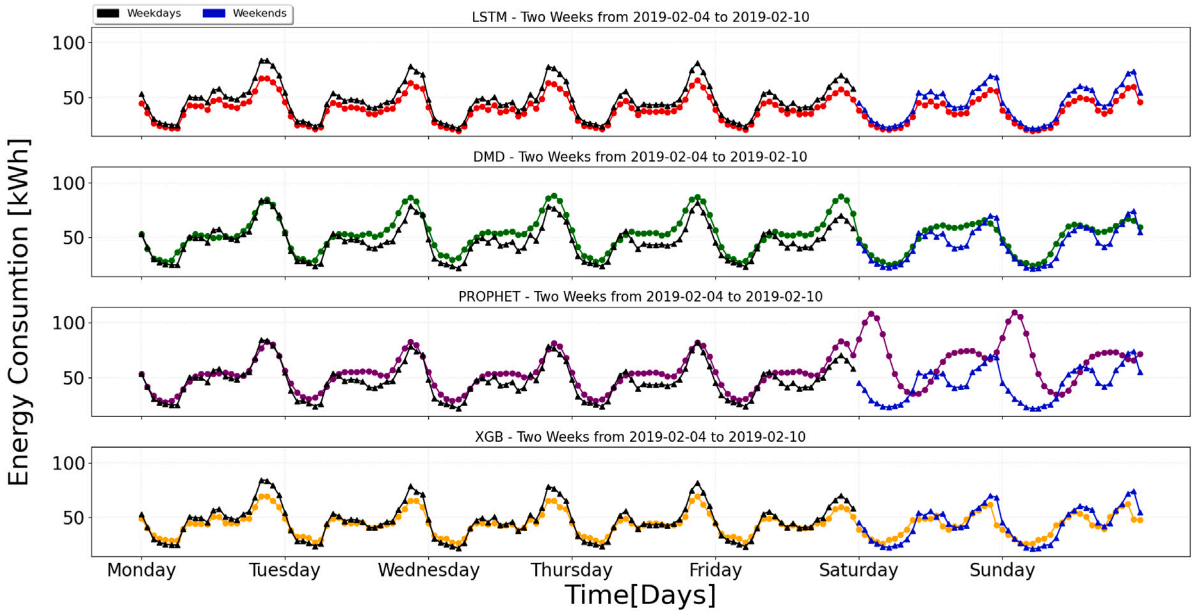

Fig. 18. Weekday and weekend difference in SS06. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

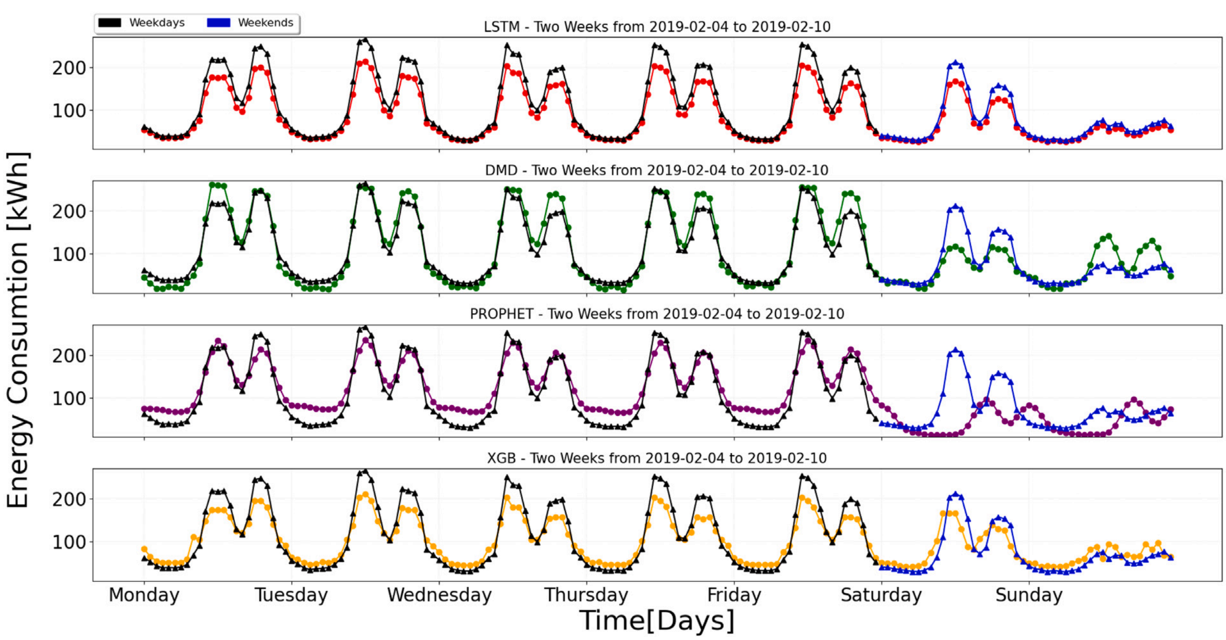

Fig. 19. Weekday and weekend difference in SS02.

The results demonstrate that DMD can remain robust without the need to split the dataset into weekdays and weekends. However, this outcome may be due to the low variation in load curves between weekdays and Saturdays. By utilizing Singular Value Decomposition (SVD), DMD creates concise data representations, leading to an efficient strategy for capturing system dynamics and making accurate predictions. In summary, the use of compact data through SVD by DMD appears to be a promising method for reliable forecasting in diverse fields.

5. Conclusions

This paper evaluates the performance of the DMD algorithm in different time horizons and different time series datasets and compares the results with those of other models. After examining three scenarios, referring to the Case Study found in Sections 4.2, 4.3 and 4.4 and comparing the DMD algorithm's predictions with those from other models, it is evident that this algorithm exhibits several advantages in the mid and long-term horizon.

The main advantages discovered in this paper on DMD over other ML and DL models are the following listed:

- **Short and long term pattern capturing**: The first part of the study shows that DMD algorithm has a remarkable ability to analyze and capture short and long term temporal patterns. Furthermore, this algorithm can identify growth and decline trends in the future, giving it an advantage over other models. This feature makes it a valuable tool for network operation and investment planning in the distant future, which is critical for effective network optimization and security. In addition, the speed and

computational efficiency of this algorithm make it ideal for processing big data applications.
- **Comparison with other models**: The comparison between DMD model shows similar performance metrics (RMSE, MAE, MAPE & MSE) in the short term when compared to more classical models. Moreover, the algorithm is capable of maintaining low metrics over a longer period of time. Compared with PROPHET, DMD always shows significantly better pattern capture and better metric performance over all the substations studied. Comparing DMD with XGBoosting regressor, the latter can show slightly better error metrics in the test for short-term forecasting. Nevertheless, its predictions can be out range for longer time horizons. Comparing DMD with LSTM, the obtained error metrics are similar, but LSTM is more computationally intensive than DMD.
- **Low computational needs**: The training procedure for a DMD model is nearly a 75% much faster than LSTM model, and compared to XGBoost, a 53% computationally less expensive.

In summary, DMD algorithm excels in capturing trends beyond conventional prediction ranges and in lengthy horizons, showcasing superior trend analysis. Furthermore, its computational efficiency surpasses that of competing models, allowing for the processing of substantial datasets for precise predictions. The algorithm's consistently strong performance metrics across all horizons make it an excellent choice for forecasting load consumption. In essence, DMD proves to be a robust tool for the analysis and prediction of intricate dynamical systems, offering valuable insights into system behavior across scientific and engineering domains. However, it is essential to acknowledge its limitations, particularly in scenarios with pronounced non-linearities or when dealing with obscured or limited data sampling. To conclude, the performance results of this study suggest the potential of DMD is in a medium and long term demand and makes it not only a feasible but an interesting alternative for forecasting applications.

Future work can compare the algorithm Dynamic Mode Decomposition with Control to other algorithms, including those that use decomposition with exogenous variables or use some average decomposition of past weather data if the predictions are not accessible. It suggests studying and comparing the performance of these algorithms with the addition of exogenous variables to gain insights and possible advantages.

CRediT authorship contribution statement

Marc Carrillo Muñoz: Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Mónica Aragüés Peñalba:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Antonio E. Saldaña González:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Marc Carrillo reports financial support, administrative support, and article publishing charges were provided by Polytechnic University of Catalonia. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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